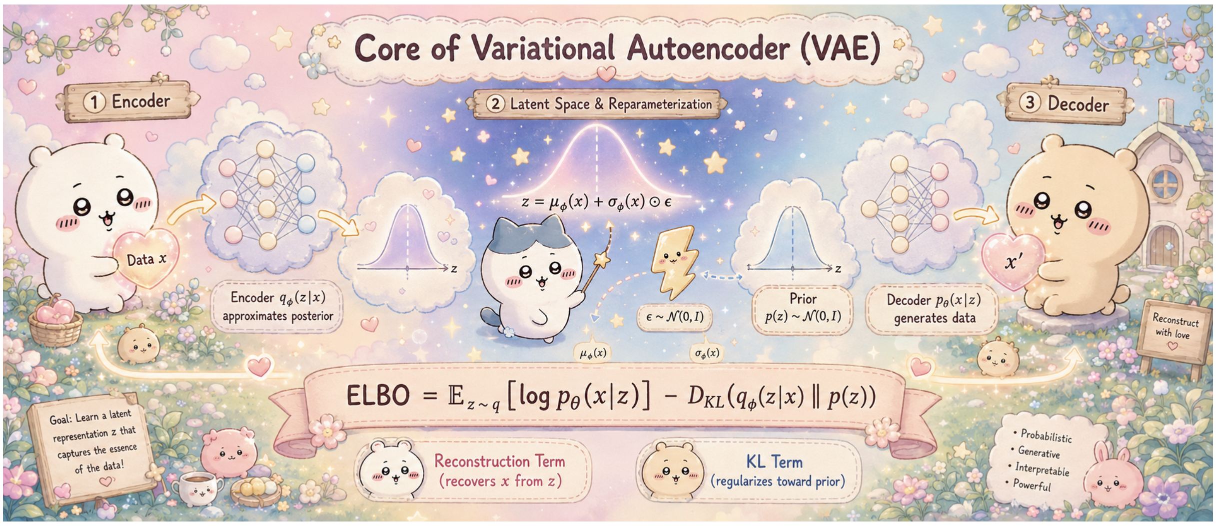
Chapter 2: Variational Autoencoder

2026 年 7 月 16 日 Deep Generative Models Notes



1 Model Structure

1.1 Encoder and Decoder

A VAE consists of two parameterized networks: encoder $q_\phi(z \mid \mathbf{x})$ and decoder $p_\theta(\mathbf{x} \mid z)$.

Definition 1: Variational Autoencoder

A **variational autoencoder** is a latent-variable generative model specified by

- a fixed **prior** $p(z)$ over a low-dimensional latent variable $z \in \mathbb{R}^d$;
- a **decoder** (generative model) $p_\theta(\mathbf{x} \mid z)$, a probabilistic map $\mathbb{R}^d \to \mathbb{R}^p$ parameterized by a neural network with weights θ;
- an **encoder** (inference model) $q_\phi(z \mid \mathbf{x})$, a probabilistic map $\mathbb{R}^p \to \mathbb{R}^d$ parameterized by a neural network with weights ϕ, used to approximate the intractable posterior $p_\theta(z \mid \mathbf{x})$.

Information flow:

$$\mathbf{x} \xrightarrow{\text{encoder}} q_\phi(z \mid \mathbf{x}) \longrightarrow z \xrightarrow{\text{decoder}} p_\theta(\mathbf{x} \mid z) \longrightarrow \mathbf{x}'$$

Remark 2: From Deterministic to Probabilistic Autoencoders

A classical autoencoder learns a deterministic map $\mathbf{x} \mapsto z \mapsto \mathbf{x}'$ for reconstruction. A VAE instead makes the latent representation probabilistic and ties it to a prior $p(z)$, so new samples are generated by drawing $z \sim p(z)$ and decoding.

1.2 Variables

- $\mathbf{x}$: observed variable
- z: latent variable with prior $p_{\text{prior}}(z) = \mathcal{N}(z; \mathbf{0}, \mathbf{I})$
- Generation: sample $z \sim p_{\text{prior}}$, then $\mathbf{x} \sim p_\theta(\mathbf{x} \mid z)$

1.3 Marginal Likelihood

$$p_\theta(\mathbf{x}) = \int p_\theta(\mathbf{x} \mid z)\, p(z)\, dz = \mathop{\mathbb{E}}_{z \sim p(z)} [p_\theta(\mathbf{x} \mid z)] .$$

This integral is intractable because $p_\theta(\mathbf{x} \mid z)$ is a nonlinear neural network, making the integral impossible to evaluate analytically.

2 Why We Need the Encoder

By Bayes' rule:

$$p_\theta(z \mid \mathbf{x}) = \frac{p_\theta(\mathbf{x} \mid z)\, p(z)}{p_\theta(\mathbf{x})}$$

Since $p_\theta(\mathbf{x})$ is intractable, so is the true posterior. We introduce a variational inference network:

$$q_\phi(z \mid \mathbf{x}) \approx p_\theta(z \mid \mathbf{x})$$

2.1 An Importance-Sampling Viewpoint

The encoder can be understood as a learned **proposal distribution** for importance sampling. Rewriting the marginal likelihood with an arbitrary $q_\phi(z \mid \mathbf{x}) > 0$,

$$p_\theta(\mathbf{x}) = \int q_\phi(z \mid \mathbf{x})\, \frac{p_\theta(\mathbf{x} \mid z)\, p(z)}{q_\phi(z \mid \mathbf{x})}\, dz = \mathop{\mathbb{E}}_{z \sim q_\phi(z|\mathbf{x})} \left[\frac{p_\theta(\mathbf{x} \mid z)\, p(z)}{q_\phi(z \mid \mathbf{x})} \right] .$$

Prior sampling is inefficient because most sampled z contribute little to explaining a fixed $\mathbf{x}$. The ideal proposal is the true posterior $p_\theta(z \mid \mathbf{x})$: then the importance weight $p_\theta(\mathbf{x} \mid z)\, p(z) / p_\theta(z \mid \mathbf{x}) = p_\theta(\mathbf{x})$ is a constant, so the estimator has zero variance and is maximally efficient. Since the true posterior is intractable, we learn $q_\phi(z \mid \mathbf{x})$ as a tractable approximation.

3 Training Objective: ELBO

3.1 Lower Bound

Definition 3: Evidence Lower Bound (ELBO)

For any datapoint $\mathbf{x}$ and any encoder $q_\phi(z \mid \mathbf{x})$ with the same support as the joint $p_\theta(\mathbf{x}, z)$, the **evidence lower bound** is

$$\mathcal{L}_{\text{ELBO}}(\theta, \phi; \mathbf{x}) \triangleq \mathop{\mathbb{E}}_{z \sim q_\phi(z|\mathbf{x})} \left[\log \frac{p_\theta(\mathbf{x}, z)}{q_\phi(z \mid \mathbf{x})} \right] = \mathop{\mathbb{E}}_{z \sim q_\phi} [\log p_\theta(\mathbf{x} \mid z)] - D_{\text{KL}}(q_\phi(z \mid \mathbf{x}) \,\|\, p(z)) .$$

The name reflects that the marginal log-likelihood $\log p_\theta(\mathbf{x})$ is often called the *evidence*.

Theorem 4: ELBO Lower Bound

$$\log p_\theta(\mathbf{x}) \;\geq\; \mathcal{L}_{\text{ELBO}}(\theta, \phi; \mathbf{x}) \;\triangleq\; \underbrace{\mathbb{E}_{z \sim q_\phi}[\log p_\theta(\mathbf{x} \mid z)]}_{\text{reconstruction}} \;-\; \underbrace{D_{\text{KL}}(q_\phi(z \mid \mathbf{x}) \,\|\, p(z))}_{\text{latent KL}}$$

Proof. Since $\log p_\theta(\mathbf{x})$ does not depend on z, we may take an expectation over $z \sim q_\phi(z \mid \mathbf{x})$, then multiply and divide by $q_\phi(z \mid \mathbf{x})$ inside the logarithm:

$$\begin{aligned}
\log p_\theta(\mathbf{x}) &= \mathbb{E}_{z \sim q_\phi}[\log p_\theta(\mathbf{x})] = \mathbb{E}_{z \sim q_\phi}\left[\log \frac{p_\theta(\mathbf{x}, z)}{p_\theta(z \mid \mathbf{x})}\right] \\
&= \mathbb{E}_{z \sim q_\phi}\left[\log \frac{p_\theta(\mathbf{x}, z)}{q_\phi(z \mid \mathbf{x})}\right] + \underbrace{\mathbb{E}_{z \sim q_\phi}\left[\log \frac{q_\phi(z \mid \mathbf{x})}{p_\theta(z \mid \mathbf{x})}\right]}_{= D_{\text{KL}}(q_\phi(z|\mathbf{x}) \,\|\, p_\theta(z|\mathbf{x})) \geq 0} \\
&\geq \mathbb{E}_{z \sim q_\phi}\left[\log \frac{p_\theta(\mathbf{x}, z)}{q_\phi(z \mid \mathbf{x})}\right] = \mathbb{E}_z[\log p_\theta(\mathbf{x} \mid z)] - D_{\text{KL}}(q_\phi(z \mid \mathbf{x}) \,\|\, p(z)),
\end{aligned}$$

where the last line uses $p_\theta(\mathbf{x}, z) = p_\theta(\mathbf{x} \mid z)\, p(z)$. The same bound follows directly from Jensen's inequality applied to the concave log:

$$\log p_\theta(\mathbf{x}) = \log \mathbb{E}_{z \sim q_\phi}\left[\frac{p_\theta(\mathbf{x}, z)}{q_\phi(z \mid \mathbf{x})}\right] \geq \mathbb{E}_{z \sim q_\phi}\left[\log \frac{p_\theta(\mathbf{x}, z)}{q_\phi(z \mid \mathbf{x})}\right].$$

□

Remark 5: Tightness of the Bound

The ELBO gap is

$$\log p_\theta(\mathbf{x}) - \mathcal{L}_{\text{ELBO}}(\theta, \phi; \mathbf{x}) = D_{\text{KL}}(q_\phi(z \mid \mathbf{x}) \,\|\, p_\theta(z \mid \mathbf{x})) \;\geq\; 0.$$

The bound is tight iff $q_\phi(z \mid \mathbf{x}) = p_\theta(z \mid \mathbf{x})$ almost everywhere.

3.2 Interpretation

- **Reconstruction term**: encourages recovery of $\mathbf{x}$ from z
- **KL term**: regularizes $q_\phi(z \mid \mathbf{x})$ toward the prior p_{prior}

4 Information-Theoretic View of ELBO

4.1 Two Joint Distributions

- **Generative joint**: $p_\theta(\mathbf{x}, z) = p(z)\, p_\theta(\mathbf{x} \mid z)$
- **Inference joint**: $q_\phi(\mathbf{x}, z) = p_{\text{data}}(\mathbf{x})\, q_\phi(z \mid \mathbf{x})$

Proposition 6: Modeling Error Bound

$$D_{\text{KL}}(p_{\text{data}}(\mathbf{x}) \,\|\, p_\theta(\mathbf{x})) \;\leq\; D_{\text{KL}}(q_\phi(\mathbf{x}, z) \,\|\, p_\theta(\mathbf{x}, z))$$

Proof. Factor both joints by the chain rule, $q_\phi(\mathbf{x}, z) = p_{\text{data}}(\mathbf{x})\, q_\phi(z \mid \mathbf{x})$ and $p_\theta(\mathbf{x}, z) =$

$p_\theta(\mathbf{x})\, p_\theta(z \mid \mathbf{x})$, and split the logarithm of the ratio:

$$\begin{aligned} D_{\mathrm{KL}}(q_\phi(\mathbf{x}, z) \,\|\, p_\theta(\mathbf{x}, z)) &= \mathop{\mathbb{E}}_{q_\phi(\mathbf{x},z)}\left[\log \frac{p_{\mathrm{data}}(\mathbf{x})\, q_\phi(z \mid \mathbf{x})}{p_\theta(\mathbf{x})\, p_\theta(z \mid \mathbf{x})}\right] \\ &= \mathop{\mathbb{E}}_{q_\phi(\mathbf{x},z)}\left[\log \frac{p_{\mathrm{data}}(\mathbf{x})}{p_\theta(\mathbf{x})}\right] + \mathop{\mathbb{E}}_{q_\phi(\mathbf{x},z)}\left[\log \frac{q_\phi(z \mid \mathbf{x})}{p_\theta(z \mid \mathbf{x})}\right]. \end{aligned}$$

The first term depends only on $\mathbf{x}$, whose marginal under $q_\phi(\mathbf{x}, z)$ is $p_{\mathrm{data}}(\mathbf{x})$, so it equals $D_{\mathrm{KL}}(p_{\mathrm{data}} \,\|\, p_\theta)$. In the second term, taking the inner expectation over $z \sim q_\phi(z \mid \mathbf{x})$ first turns each fiber into a KL divergence, leaving an outer expectation over $\mathbf{x} \sim p_{\mathrm{data}}$:

$$\begin{aligned} D_{\mathrm{KL}}(q_\phi(\mathbf{x}, z) \,\|\, p_\theta(\mathbf{x}, z)) &= D_{\mathrm{KL}}(p_{\mathrm{data}} \,\|\, p_\theta) + \mathop{\mathbb{E}}_{p_{\mathrm{data}}}\left[D_{\mathrm{KL}}(q_\phi(z \mid \mathbf{x}) \,\|\, p_\theta(z \mid \mathbf{x}))\right] \\ &\geq D_{\mathrm{KL}}(p_{\mathrm{data}} \,\|\, p_\theta)\,, \end{aligned}$$

since the second term is an expectation of nonnegative KL divergences. □

4.2 Error Decomposition

- $D_{\mathrm{KL}}(p_{\mathrm{data}} \,\|\, p_\theta)$: **modeling error**
- $\mathbb{E}_{p_{\mathrm{data}}}[D_{\mathrm{KL}}(q_\phi(z \mid \mathbf{x}) \,\|\, p_\theta(z \mid \mathbf{x}))]$: **inference error**

Remark 7: ELBO and Inference Error

$$\log p_\theta(\mathbf{x}) - \mathcal{L}_{\mathrm{ELBO}}(\theta, \phi; \mathbf{x}) = D_{\mathrm{KL}}(q_\phi(z \mid \mathbf{x}) \,\|\, p_\theta(z \mid \mathbf{x})) \geq 0$$

Maximizing ELBO therefore reduces inference error.

5 Reparameterization and Gradient Estimation

5.1 The Gradient Problem

Training maximizes $\mathbb{E}_{p_{\mathrm{data}}(\mathbf{x})}[\mathcal{L}_{\mathrm{ELBO}}(\theta, \phi; \mathbf{x})]$ by stochastic gradient ascent. The gradient with respect to the decoder θ is straightforward, since the sampling distribution q_ϕ does not depend on θ:

$$\nabla_\theta \mathop{\mathbb{E}}_{z \sim q_\phi}[\log p_\theta(\mathbf{x} \mid z)] = \mathop{\mathbb{E}}_{z \sim q_\phi}[\nabla_\theta \log p_\theta(\mathbf{x} \mid z)]\,.$$

The encoder gradient ∇_ϕ is harder because the sampling distribution itself depends on ϕ:

$$\nabla_\phi \mathop{\mathbb{E}}_{z \sim q_\phi(z|\mathbf{x})}[f(z)] \neq \mathop{\mathbb{E}}_{z \sim q_\phi(z|\mathbf{x})}[\nabla_\phi f(z)]\,.$$

5.2 The Reparameterization Trick

Definition 8: Reparameterization

A distribution $q_\phi(z \mid \mathbf{x})$ is **reparameterizable** if there exist a fixed base distribution $p(\boldsymbol{\epsilon})$ (independent of ϕ) and a differentiable map $g_\phi(\boldsymbol{\epsilon}, \mathbf{x})$ such that

$$z \sim q_\phi(z \mid \mathbf{x}) \quad \iff \quad z = g_\phi(\boldsymbol{\epsilon}, \mathbf{x}), \quad \boldsymbol{\epsilon} \sim p(\boldsymbol{\epsilon}).$$

For the diagonal Gaussian encoder $q_\phi(z \mid \mathbf{x}) = \mathcal{N}\Big(z; \boldsymbol{\mu}_\phi(\mathbf{x}), \mathrm{diag}(\boldsymbol{\sigma}^2_\phi(\mathbf{x}))\Big)$ this is

$$z = \boldsymbol{\mu}_\phi(\mathbf{x}) + \boldsymbol{\sigma}_\phi(\mathbf{x}) \odot \boldsymbol{\epsilon}, \qquad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}),$$

where $\odot$ denotes the elementwise product.

Proposition 9: Pathwise Gradient Estimator

If $q_\phi(z \mid \mathbf{x})$ is reparameterizable via $z = g_\phi(\boldsymbol{\epsilon}, \mathbf{x})$, then for any differentiable f,

$$\nabla_\phi \mathop{\mathbb{E}}_{z\sim q_\phi(z|\mathbf{x})}[f(z)] = \mathop{\mathbb{E}}_{\boldsymbol{\epsilon}\sim p(\boldsymbol{\epsilon})}[\nabla_\phi f(g_\phi(\boldsymbol{\epsilon}, \mathbf{x}))] \,.$$

Proof. Because $p(\boldsymbol{\epsilon})$ does not depend on ϕ, the expectation operator and the gradient commute under the change of variables $z = g_\phi(\boldsymbol{\epsilon}, \mathbf{x})$:

$$\nabla_\phi \mathop{\mathbb{E}}_{z\sim q_\phi(z|\mathbf{x})}[f(z)] = \nabla_\phi \mathop{\mathbb{E}}_{\boldsymbol{\epsilon}\sim p(\boldsymbol{\epsilon})}[f(g_\phi(\boldsymbol{\epsilon}, \mathbf{x}))] = \mathop{\mathbb{E}}_{\boldsymbol{\epsilon}\sim p(\boldsymbol{\epsilon})}[\nabla_\phi f(g_\phi(\boldsymbol{\epsilon}, \mathbf{x}))] \,,$$

where the last step exchanges ∇_ϕ with the (now ϕ-free) expectation. The chain rule $\nabla_\phi f = (\partial_z f)\,(\partial_\phi g_\phi)$ then propagates the gradient through the sampled z. $\square$

5.3 Monte Carlo ELBO and Its Gradient

Combining reparameterization with a single Monte Carlo sample per datapoint gives the unbiased estimator actually used in practice. With $z = g_\phi(\boldsymbol{\epsilon}, \mathbf{x})$ and $\boldsymbol{\epsilon} \sim p(\boldsymbol{\epsilon})$,

$$\widehat{\mathcal{L}_{\mathrm{ELBO}}}(\theta, \phi; \mathbf{x}) = \log p_\theta(\mathbf{x} \mid g_\phi(\boldsymbol{\epsilon}, \mathbf{x})) - D_{\mathrm{KL}}(q_\phi(z \mid \mathbf{x}) \,\|\, p(z)) \,,$$

and its gradient (when the KL term is available in closed form, see Section 6) is

$$\nabla_{\theta,\phi} \widehat{\mathcal{L}_{\mathrm{ELBO}}}(\theta, \phi; \mathbf{x}) = \nabla_{\theta,\phi}[\log p_\theta(\mathbf{x} \mid g_\phi(\boldsymbol{\epsilon}, \mathbf{x})) - D_{\mathrm{KL}}(q_\phi(z \mid \mathbf{x}) \,\|\, p(z))] \,,$$

which is an unbiased estimator of $\nabla_{\theta,\phi} \mathcal{L}_{\mathrm{ELBO}}(\theta, \phi; \mathbf{x})$ and is computed by ordinary backpropagation.

Remark 10: Why Not the Score-Function Estimator

The general-purpose score-function (REINFORCE) estimator

$$\nabla_\phi \mathop{\mathbb{E}}_{z\sim q_\phi}[f(z)] = \mathop{\mathbb{E}}_{z\sim q_\phi}[f(z)\, \nabla_\phi \log q_\phi(z \mid \mathbf{x})]$$

is unbiased and does not require reparameterization, but often has much higher variance. For continuous latents, pathwise gradients are typically more stable in practice.

6 Gaussian VAE

6.1 Parameterization

Encoder (diagonal Gaussian):

$$q_\phi(z \mid \mathbf{x}) \triangleq \mathcal{N}\big(z;\; \boldsymbol{\mu}_\phi(\mathbf{x}),\; \mathrm{diag}\big(\boldsymbol{\sigma}^2_\phi(\mathbf{x})\big)\big)$$

where $\boldsymbol{\mu}_\phi : \mathbb{R}^p \to \mathbb{R}^d$, $\boldsymbol{\sigma}_\phi : \mathbb{R}^p \to \mathbb{R}^d_+$.

Decoder (isotropic Gaussian):

$$p_\theta(\mathbf{x} \mid z) \triangleq \mathcal{N}\big(\mathbf{x};\; \mathbf{M}_\theta(z),\; \sigma^2 \mathbf{I}\big)$$

where $\mathbf{M}_\theta : \mathbb{R}^d \to \mathbb{R}^p$ and $\sigma > 0$ is a fixed small constant.

6.2 Equivalent Optimization

Expanding the reconstruction term:

$$\mathbb{E}_z[\log p_\theta(\mathbf{x} \mid z)] = -\frac{1}{2\sigma^2}\mathbb{E}_z\left[\|\mathbf{x} - \mathbf{M}_\theta(z)\|^2\right] + \text{const}$$

So maximizing ELBO is equivalent to:

$$\min_{\theta,\phi} \mathbb{E}_{p_{\text{data}}}\left[\mathbb{E}_{q_\phi}\left[\frac{1}{2\sigma^2}\|\mathbf{x} - \mathbf{M}_\theta(z)\|^2\right] + D_{\text{KL}}(q_\phi(z \mid \mathbf{x}) \,\|\, p_{\text{prior}}(z))\right]$$

Here σ controls the reconstruction–regularization trade-off. The reconstruction term is estimated via reparameterized sampling (Section 5), while the KL term is closed form.

Proposition 11: Closed-Form Gaussian KL

When both q_ϕ and p_{prior} are Gaussian with diagonal covariance,

$$D_{\text{KL}}\big(\mathcal{N}(\boldsymbol{\mu}, \operatorname{diag}(\boldsymbol{\sigma}^2)) \,\|\, \mathcal{N}(\mathbf{0}, \mathbf{I})\big) = \frac{1}{2}\sum_{j=1}^{d}\big(\mu_j^2 + \sigma_j^2 - \log\sigma_j^2 - 1\big)\,.$$

Proof. Because both distributions factorize over the d coordinates, the KL divergence is the sum of the per-coordinate divergences; it suffices to prove the one-dimensional case for $q = \mathcal{N}(\mu, \sigma^2)$ and $p = \mathcal{N}(0, 1)$. Writing out the densities,

$$\log\frac{q(z)}{p(z)} = \log\frac{1}{\sigma} - \frac{(z-\mu)^2}{2\sigma^2} + \frac{z^2}{2} = -\frac{1}{2}\log\sigma^2 - \frac{(z-\mu)^2}{2\sigma^2} + \frac{z^2}{2}.$$

Taking the expectation over $z \sim q$ and using $\mathbb{E}_q[(z-\mu)^2] = \sigma^2$ together with $\mathbb{E}_q[z^2] = \mu^2 + \sigma^2$,

$$D_{\text{KL}}(q \,\|\, p) = \mathbb{E}_q\left[\log\frac{q(z)}{p(z)}\right] = -\frac{1}{2}\log\sigma^2 - \frac{1}{2} + \frac{\mu^2+\sigma^2}{2} = \frac{1}{2}\big(\mu^2 + \sigma^2 - \log\sigma^2 - 1\big)\,.$$

Summing over the d independent coordinates yields the stated formula. □

Remark 12: A Fully Differentiable Objective

With reparameterized reconstruction and closed-form KL (Proposition 11), the per-sample objective is fully differentiable in (θ, ϕ) and optimized by standard backpropagation.

7 Hierarchical VAE (HVAE)

7.1 Motivation

Hierarchical VAEs introduce a hierarchy of latent variables, allowing the model to capture features at multiple abstraction levels.

7.2 Hierarchical Encoder-Decoder Chain

$$\mathbf{x} \underset{q_\phi(z_1|\mathbf{x})}{\xrightarrow{p_\theta(\mathbf{x}|z_1)}} z_1 \underset{q_\phi(z_2|z_1)}{\xrightarrow{p_\theta(z_1|z_2)}} \cdots \underset{q_\phi(z_L|z_{L-1})}{\xrightarrow{p_\theta(z_{L-1}|z_L)}} z_L$$

7.3 Modeling

The generative joint distribution is:

$$p_\theta(\mathbf{x}, z_{1:L}) = p_\theta(\mathbf{x} \mid z_1)\prod_{i=2}^{L} p_\theta(z_{i-1} \mid z_i)\, p(z_L)$$

The marginal data distribution becomes:

$$p_{\text{HVAE}}(\mathbf{x}) = \int p_\theta(\mathbf{x}, z_{1:L})\, dz_{1:L}$$

7.4 Generation and Encoding

Generation samples top latent $z_L \sim p(z_L)$, then decodes downward layer by layer to z_1, and finally generates $\mathbf{x}$ from $p_\theta(\mathbf{x} \mid z_1)$.

Encoding follows a Markov factorization:

$$q_\phi(z_{1:L} \mid \mathbf{x}) = q_\phi(z_1 \mid \mathbf{x}) \prod_{i=2}^{L} q_\phi(z_i \mid z_{i-1})$$

This bottom-up Markov chain mirrors the top-down decoder and keeps inference/sampling tractable through simple conditional factors.

8 HVAE ELBO

8.1 Derivation

$$\begin{aligned}
\log p_{\text{HVAE}}(\mathbf{x}) &= \log \int p_\theta(\mathbf{x}, z_{1:L})\, dz_{1:L} \\
&= \log \int q_\phi(z_{1:L} \mid \mathbf{x}) \frac{p_\theta(\mathbf{x}, z_{1:L})}{q_\phi(z_{1:L} \mid \mathbf{x})}\, dz_{1:L} \\
&= \log \mathop{\mathbb{E}}_{q_\phi(z_{1:L}|\mathbf{x})} \left[\frac{p_\theta(\mathbf{x}, z_{1:L})}{q_\phi(z_{1:L} \mid \mathbf{x})} \right] \\
&\geq \mathop{\mathbb{E}}_{q_\phi(z_{1:L}|\mathbf{x})} \left[\log \frac{p_\theta(\mathbf{x}, z_{1:L})}{q_\phi(z_{1:L} \mid \mathbf{x})} \right] \triangleq \mathcal{L}_{\text{ELBO}}^{\text{HVAE}}(\theta, \phi; \mathbf{x})
\end{aligned}$$

8.2 Three-Term Decomposition

Substituting the generative and inference factorizations into $\log \frac{p_\theta(\mathbf{x}, z_{1:L})}{q_\phi(z_{1:L}|\mathbf{x})}$ and using $z_0 \equiv \mathbf{x}$ yields a reconstruction term, adjacent-layer KL matching terms, and a top-level KL to the prior:

$$\begin{aligned}
\mathcal{L}_{\text{ELBO}}^{\text{HVAE}}(\theta, \phi; \mathbf{x}) = &\underbrace{\mathop{\mathbb{E}}_{q_\phi(z_1|\mathbf{x})} [\log p_\theta(\mathbf{x} \mid z_1)]}_{\text{reconstruction term}} \\
&- \underbrace{\sum_{i=1}^{L-1} \mathop{\mathbb{E}}_{q_\phi(z_{1:L}|\mathbf{x})} [D_{\text{KL}}(q_\phi(z_i \mid z_{i-1}) \,\|\, p_\theta(z_i \mid z_{i+1}))]}_{\text{adjacent-layer matching terms}} \\
&- \underbrace{\mathop{\mathbb{E}}_{q_\phi(z_{L-1}|\mathbf{x})} [D_{\text{KL}}(q_\phi(z_L \mid z_{L-1}) \,\|\, p(z_L))]}_{\text{top-level KL regularizer}}
\end{aligned}$$

where $z_0 \equiv \mathbf{x}$ for notational convenience.

9 Pathologies of Deep Networks in Flat VAE

9.1 Reason 1: Deeper Networks Do Not Expand the Variational Family

In the standard variational family

$$q_\phi(z \mid \mathbf{x}) = \mathcal{N}\left(z; \mu_\phi(\mathbf{x}), \operatorname{diag}(\sigma_\phi^2(\mathbf{x}))\right),$$

deeper encoder networks may improve parameter fitting quality, but they do not change the family itself. If the true posterior $p_\theta(z \mid \mathbf{x})$ is multi-modal, a diagonal Gaussian approximation remains fundamentally limited.

9.2 Reason 2: Posterior Collapse

The expected ELBO over data can be decomposed as:

$$\mathop{\mathbb{E}}_{p_{\text{data}}(\mathbf{x})}[\mathcal{L}_{\text{ELBO}}(\mathbf{x})] = \mathop{\mathbb{E}}_{p_{\text{data}}(\mathbf{x})\, q_\phi(z|\mathbf{x})}[\log p_\theta(\mathbf{x} \mid z)] - I_\phi(\mathbf{x}; z) - D_{\text{KL}}(\bar{q}_\phi(z) \,\|\, p(z)) \,,$$

where the aggregated posterior is

$$\bar{q}_\phi(z) \triangleq \mathop{\mathbb{E}}_{p_{\text{data}}(\mathbf{x})}[q_\phi(z \mid \mathbf{x})],$$

and the mutual information is

$$I_\phi(\mathbf{x}; z) \triangleq \mathop{\mathbb{E}}_{p_{\text{data}}(\mathbf{x})}[D_{\text{KL}}(q_\phi(z \mid \mathbf{x}) \,\|\, \bar{q}_\phi(z))] \,.$$

This decomposition follows by averaging the per-datapoint latent KL and inserting the aggregated posterior $\bar{q}_\phi(z)$ inside the logarithm:

$$\begin{aligned}
\mathop{\mathbb{E}}_{p_{\text{data}}(\mathbf{x})}[D_{\text{KL}}(q_\phi(z \mid \mathbf{x}) \,\|\, p(z))] &= \mathop{\mathbb{E}}_{p_{\text{data}}(\mathbf{x})\, q_\phi(z|\mathbf{x})}\left[\log \frac{q_\phi(z \mid \mathbf{x})}{\bar{q}_\phi(z)} + \log \frac{\bar{q}_\phi(z)}{p(z)}\right] \\
&= \underbrace{\mathop{\mathbb{E}}_{p_{\text{data}}(\mathbf{x})}[D_{\text{KL}}(q_\phi(z \mid \mathbf{x}) \,\|\, \bar{q}_\phi(z))]}_{=I_\phi(\mathbf{x};z)} + \underbrace{\mathop{\mathbb{E}}_{\bar{q}_\phi(z)}\left[\log \frac{\bar{q}_\phi(z)}{p(z)}\right]}_{=D_{\text{KL}}(\bar{q}_\phi(z) \,\|\, p(z))},
\end{aligned}$$

where the second term uses that the marginal of z under $p_{\text{data}}(\mathbf{x})\, q_\phi(z \mid \mathbf{x})$ is exactly $\bar{q}_\phi(z)$. Substituting back into $\mathbb{E}_{p_{\text{data}}}[\mathcal{L}_{\text{ELBO}}] = \mathbb{E}_{p_{\text{data}}\, q_\phi}[\log p_\theta(\mathbf{x} \mid z)] - \mathbb{E}_{p_{\text{data}}}[D_{\text{KL}}(q_\phi(z \mid \mathbf{x}) \,\|\, p(z))]$ gives the three-term form above.

If the decoder is too expressive, the model class may contain solutions with $p_\theta(\mathbf{x} \mid z) \approx r(\mathbf{x}) \approx p_{\text{data}}(\mathbf{x})$ that almost ignore z. Then $q_\phi(z \mid \mathbf{x})$ is pushed toward $p(z)$, implying $I_\phi(\mathbf{x}; z) \to 0$ and KL penalties close to zero.

Remark 13: Consequences of Posterior Collapse

- z becomes nearly independent of $\mathbf{x}$, carrying little useful information.
- Conditioning on z or moving in latent space has little effect on generated samples.

9.3 What Hierarchy Changes?

One structural response to posterior collapse is to adopt hierarchical latents (HVAE; Section 7). Instead of routing all information through a single code, the model distributes representation across multiple levels, with adjacent-layer matching terms that constrain inference and generation at every interface.

The HVAE ELBO (Section 8) can be written in log-difference form as

$$\begin{aligned}
\mathcal{L}_{\text{ELBO}}^{\text{HVAE}}(\theta, \phi; \mathbf{x}) = &\mathop{\mathbb{E}}_{q_\phi(z_1|\mathbf{x})}[\log p_\theta(\mathbf{x} \mid z_1)] \\
&- \mathop{\mathbb{E}}_{q_\phi(z_{1:2}|\mathbf{x})}[\log q_\phi(z_1 \mid \mathbf{x}) - \log p_\theta(z_1 \mid z_2)] \\
&- \sum_{i=2}^{L-1} \mathop{\mathbb{E}}_{q_\phi(z_{1:i+1}|\mathbf{x})}[\log q_\phi(z_i \mid z_{i-1}) - \log p_\theta(z_i \mid z_{i+1})] \\
&- \mathop{\mathbb{E}}_{q_\phi(z_{L-1}|\mathbf{x})}[D_{\text{KL}}(q_\phi(z_L \mid z_{L-1}) \,\|\, p(z_L))] \,.
\end{aligned}$$

Under $z_0 \equiv \mathbf{x}$, each log-difference term is exactly the corresponding adjacent-layer KL matching term in Section 8; the top-level term is already a KL.

Remark 14: Layer-Wise Abstraction in HVAE

Each latent level interacts only with its neighbors: the encoder passes information upward through $q_\phi(z_i \mid z_{i-1})$, and the decoder passes information downward through $p_\theta(z_{i-1} \mid z_i)$. These properties stem from the hierarchical latent graph, not from simply deepening networks in a flat VAE. Because matching penalties act at every layer, a powerful decoder cannot ignore the entire latent hierarchy without incurring multiple interface costs; HVAE is therefore typically less collapse-prone than a flat VAE with a deeper encoder alone.